

The fidelity-utility paradox and its resolution

Surrogate (fidelity)

Action response surface

PPO policy

Live BOPTEST transfer

v3 black-box · 1 h step

RMSE 1.557 °C

smooth · monotone

roughness 1.0×

modulating

24% saturation

stable transfer

$m_s \approx 0.08$

v3.5 / matched v3

RMSE 0.644/0.876 °C

rough · non-monotone

7.9–9.4× rougher

near bang-bang

100% saturation

closed-loop collapse

$m_s > 1$, >77% viol.

hybrid: v3 rollout

+ frozen v3.5 sensor

smooth (v3) +

plausibility sensor

modulating

25% saturation

robust transfer

$m_s = 0.041$, viol <5%